



ME 609: Functional Materials

Course Code	ME 609
Course Name	Functional Materials
L-T-P-C	3-0-0-3
Intended for	B. Tech
Distribution	Elective
Approval	2nd Adhoc Senate (May 2012)

Course Contents

- **Introduction**
Definition of functional materials, Different kind of functional materials; Use of functionalities of materials in fabricating devices, Causes for observed functionality in a material; Functionality arising due to (i) electronic, (ii) spin, and (iii) ionic degrees of freedom; Exploitation of combined effects in designing new functional materials. [3 Lectures]
- **Functionality driven by electronic degrees of freedom**
Atoms and crystalline solids; electronic states of atoms and crystalline solids; Formation of bands in crystalline solids; Band dispersions; Density of states; Metals, semiconductors and insulators; Direct and indirect band gap semiconductors; Formation of impurity bands in the p -type and n -type semiconductors; Electrons effective mass in a semiconductor; Transport and optical properties of a semiconductor; Opto-electronic materials. [12 Lectures]
- **Functionality driven by spin degrees of freedom**
Formation of magnetic moment in an atom; Spin and orbital part of magnetic moment in a solid; Magnetization of a solid; Diamagnetic, paramagnetic, ferromagnetic and antiferromagnetic materials; Different kind of antiferromagnetic structures; Exchange interaction; Determination of magnetic transition temperature using mean-field theory; Formation of domain wall in ferromagnetic material; Soft and hard ferromagnets; CMR/GMR materials. [10 Lectures]
- **Functionality driven by ionic degrees of freedom**
Covalent, ionic and metallic solids; Formation of dipole moment; Polarization of a material; Paraelectric, ferroelectric, antiferroelectric, piezoelectric, and pyroelectric materials; formation of domain wall in ferroelectric material; Multiferroic materials. [5 Lectures]
- **Project**
Brief overview of density functional theory; Different kinds of exchange-correlation functional; Use of full-potential LMTO and LAPW methods in designing functional materials. [12 Lectures]

Text and Reference Books

1. N.W.Ashcroft and N.D. Mermin, **Solid State Physics**, Harcourt College Publishers
2. Marius Grundmann, **The Physics of Semiconductors: An Introduction Including Devices and Nanophysics**, Springer Berlin Heidelberg New York
3. R.M.Martin, **Electronic Structure: Basic Theory and Practical Methods**, Cambridge University Press
4. K.F. Wang, J. M. Liu, and Z.F.Ren, **Multiferroicity: the coupling between magnetic and polarization orders**, *Advances in Physics*, 58, 321 (2009)